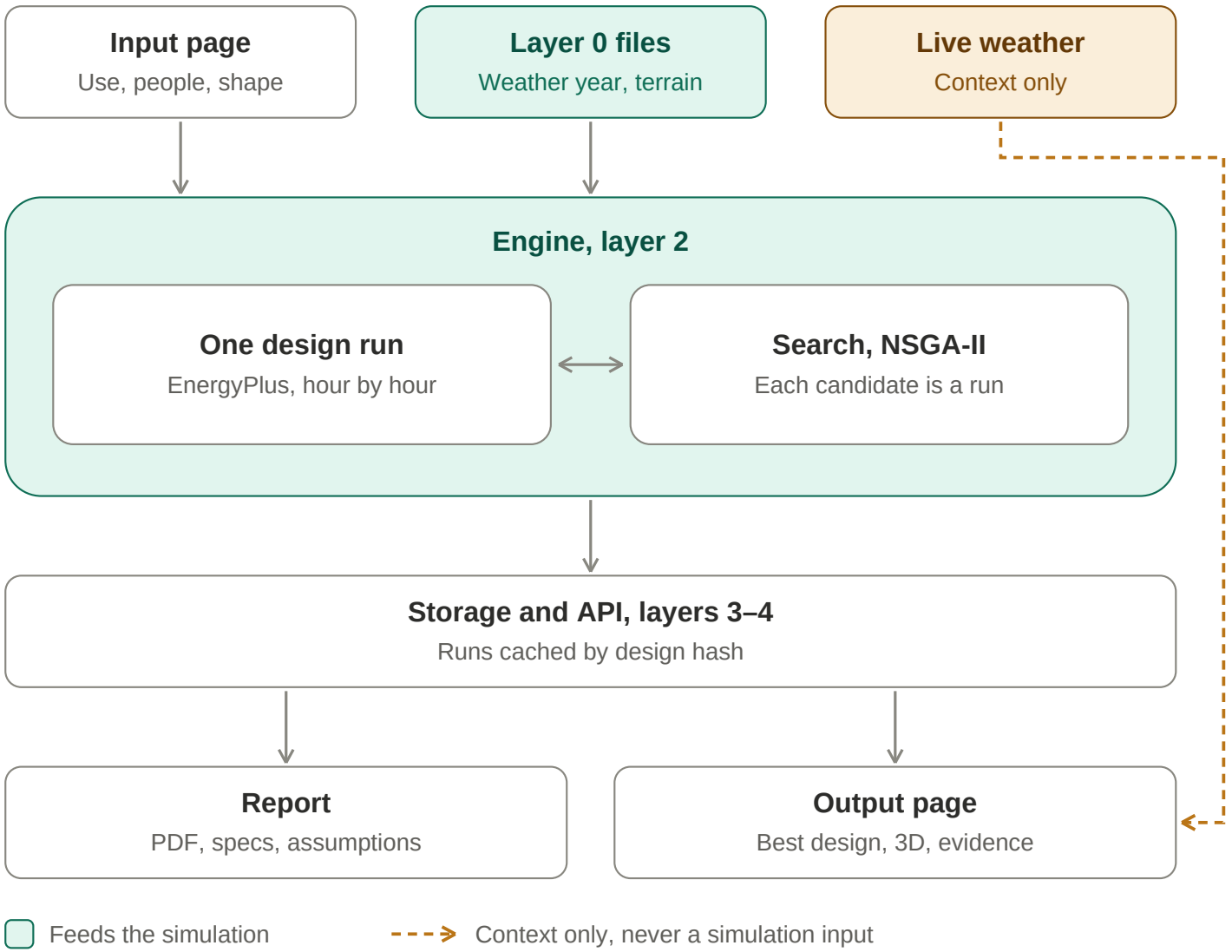
System flow

How an input becomes a recommended shelter



What each block is

Input page. Shelter use, people, floor area, orientation, arrangement, glazing, air leakage, comfort target and search size.

Layer 0 files. TMYx weather year for Leh (station 427053, 8,760 hours), SRTM terrain, 36-direction horizon, ground temperatures and 15 materials. Read from files, so it works offline.

One design run. Builds the geometry and wall build-ups from the inputs and runs EnergyPlus 24.2 against the weather year.

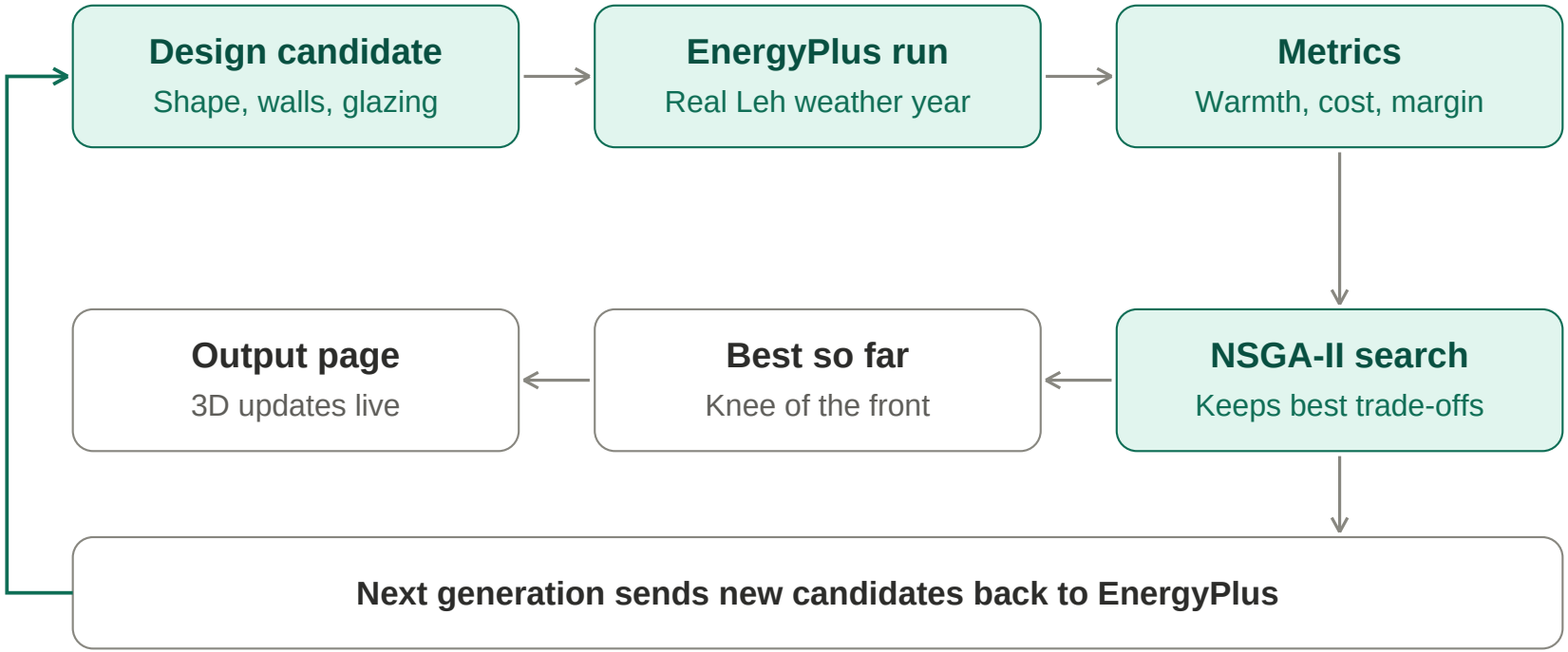
Search. NSGA-II evolves designs. Every candidate is a full EnergyPlus run, no shortcut model.

Storage and API. SQLite keeps every run, keyed by design hash, and FastAPI serves it.

Live weather. Current conditions from Open-Meteo, shown for context. The simulation never reads it.

The search loop

How the best shelter is found, one generation at a time



What the search controls

Changes: wall mass 150–550 mm · insulation 20–180 mm · south glazing 8–45 % of the wall.

Balances: warmth (degree hours outside the comfort band), cost and the worst-night margin.

Never breaks: condensation at most 2,000 hours · walls at most 900 mm · the fresh-air minimum.

Size: 100, 300 or 500 designs, about 6, 18 or 31 minutes at 3.7 seconds per design.

After a design is picked

The chosen design gets one full run that the search skips: the comparison with current construction, the 8-run uncertainty band, heating demand and diesel, condensation, cost and buildability, and generated recommendations.

Those results feed the output page and the report.

The weather year and site never change during the search. Only the design does.

Corrections to the proposed flow

Where the earlier diagram differs from how the project works

Proposed	How it actually works
A weather API fetches conditions for the solver	EnergyPlus reads the frozen TMYx weather year from Layer 0. Live weather from Open-Meteo is shown for context only and never enters a simulation, so results repeat exactly and work offline.
User-defined region	Leh, Ladakh only. Each new location needs its own weather year, terrain and horizon from Layer 0.
Material database syncs into shelter geometry	Materials are ingested once into SQLite and set the wall, roof and floor build-ups, not the shape.
Optimisation updates climate parameters	The search changes only the design: wall mass, insulation and south glazing. Weather and site stay fixed.
Project management tools receive updates	Not part of the project.

Added, because it exists in the project

- A run of current construction for every design, so each result is shown as a difference.
- An 8-run uncertainty sweep that turns headline numbers into a range.
- Storage and API layers that cache runs and serve every screen.
- The best design so far, published after every generation.
- Layer 1 validation (ASHRAE Standard 140 cases, ANSYS thermal bridges and fluid dynamics). Planned, not yet committed.